清华大学热能工程系 2003 级“燃烧学”期中考试试卷答案提示

考试时间： 2005 年 4 月 18 日 班级：_____

学号：_____ 姓名：_____

Instructions(提示):

1 考试为**开卷考试**，允许带教科书，作业，课件，笔记本和另外的参考书各一样，但不能互借，答题必须独立完成。（The examination is **OPEN BOOKS**. Calculators, textbook, homework and notebook are allowed, but they can not be lent to others. You should answer the questions independently and help is not allowed）

2 试卷共有8题，你要在120分钟内完成这次考试，先检查一下你的卷子，题目是否对，共有4页。答案必须写在答题卷上，试题卷上回答无效。答题必须明确，清晰，计算题应给出单位。（There are four questions on this exam. You have up to 120 minutes to work this exam. Please be sure that you have a complete exam package. Please work exam problems on the pages provided. Show your work neatly and write your answers. Give units for all answers requiring them.）

3 如果试卷不对，请立即与监考教师说明。（If a page is missing, take the exam to a proctor immediately.）

4 强烈推荐你先读一下所有的问题，并对回答问题的时间进行按排。答卷完成后请再仔细检查一下。（Budget your time for each question and be sure to first answer those questions that you know how to work. Check your work after completing the exam.）

5 请你现在就在答卷纸和试卷纸上写上你的**名字与学号**，并在你的答卷纸上标明你的试卷的类型（A或B卷）（**PRINT your name NOW** in the spaces at the top of your pages.）

6 答卷可以用中文或英文，如果是双语的，以中文为准。试卷有英译的，也以中文为准。(The answers can be in English or Chinese. If two languages are used in one answer, I will take Chinese as the selection. Some of the problems have been translated in English. If there are any difference between Chinese and English, Chinese is for your choice.)

问题（problems）

1 简要回答下列问题，选择或填空（每题2分， 8题，共16分）（Answer the following questions briefly）

1.1 有些反应的活化能为零，请解释其物理含义。(Sometimes the reactive energy equals zero. Please explain the reason.)

答：应该是特别易反应，只要一碰撞就发生反应，没有能量的要求。

1.2 化学平衡是指反应物与产物的浓度不变，即反应已经停止，对否？（Chemical equilibrium means the concentration of the reactants and products do not change and the reaction ceased.）

答：错，反应没有停止，只是正反平衡速度相等。

1.3 有25%过量空气时，碳的空燃比为 C kg air/kg fuel(There are 25% excess air in the combustion of carbon. The Air fuel ratio of the reaction is kg air/kg fuel)

A．9.12 B. 11.4 C. 14.25 D. 16.25

1.4 是不是温度越高，燃烧越完全？(the combustion is more complete when the combustion temperature is higher)

答：否，从热力学第二定律角度出发，结论可能是温度越高越容易出现分解。

1.5 任何反应的高位热值一定高于低位热值，对否？(the higher heating value is always higher than the lower heating value. True or false?)

答：否，不含氢的燃料燃烧，两者要等

1.6 标准参考状态由三个主要的因素来确定，它们分别是：温度（298.15K）、压力 和 元素的稳定状态 。(We define a standard reference state for three parameters. They are _____ 、_____and _____ .)

1.7 一种煤的收到基成分为M_{ar}=10%，A_{ar}=20%，V_{ar}=25%，则其干燥无灰基的挥发份V_{daf}为： B (One kind of coal consists of M_{ar}=10%，A_{ar}=20%，V_{ar}=25% with the dry base. The dry ash free volatile content is: _____)

A．31.25% B. 35.71% C. 17.5% D. 20%

1.8 为什么烟气再循环会减少火焰温度，如果循环烟气温度为火焰温度时将如何？（Why does flue-gas recirculation decrease flame temperature? What happens if the flue gas recirculated is at the flame temperature?）

答：因为一般来说烟气的温度低于火焰温度，而只起到了循环混合的作用。如果温度为火焰温度，则不变

2 （14分）试证明一级反应，在等温条件下转化率达到99.9%所需的时间为转化率达到50%的所需时间的10倍。如果反应是0级或2级时，两者的时间比为多少？When the reaction is in first order and isothermal， prove that the ratio is 10 between the time when

the conversion rate is 99.9% and the time when the conversion rate is 50%. If the reaction order is 0 or 2, what is the ratio?

答：主要注意99.9%时浓度达到初始的0.001，参看p.133页有关内容，一级反应的用式4.67，则一级反应$t_{0.999}/t_{0.5}=\ln 0.5/\ln 0.001=10.0343$.当然是10倍。

如果反应是0级：$\frac{d[A]}{dt}=-k$，则有$t_{0.999}/t_{0.5}=1.998$

如果反应是2级：则$\frac{d[A]}{dt}=-k[A]^2$，则$\frac{1}{[A]_t}=kt+\frac{1}{[A]_o}$则有$t_{0.999}/t_{0.5}=999$

3 (10分)目前在考虑用替代燃料代替汽油燃烧（汽油分子式用C_8H_{18}式表示，其低位热值为42.5MJ/kg），目前主要的替代燃料是指乙醇（分子式为C_2H_6O，其低位热值仅为27.6MJ/kg），有人断言，因为其热值低，所以乙醇的理论燃烧温度要比柴油低，无法在内燃机中使用，你认为如何？为什么？(The ethanol(C_2H_6O,LHV is 27.6MJ/kg) is considered a good substitute for gasoline（C_8H_{18}，LHV is 42.5MJ/kg）, but someone said that for the lower LHV for methanol its adiabatic flame temperature will be lower than that of the gasoline. It means ethanol can not be used. How do you think and explain the reason.)

答：这种说法不正确，由于乙醇中含氧，虽然热值低了一些，但合理的配风，还是维持合理的理论燃烧温度，同时由于氧的存在可以大大降低污染物CO和SOOT形成的可能，可以实现更低氧的燃烧，从而也可以减少NOx的形成。可以计算出如果完全燃烧的话，其理论燃烧温度变化不大的结论。

4 （5分） 一个实验员发现当她所用的氧化剂为高纯度的O2（21％），和高纯度的N2（79％）时，无法点燃预混的一氧化碳火焰。当用室内空气做氧化剂时，火焰却能持续燃烧。试给出合理解释。(A technician found that it is difficult to ignite the CO flame when the pure Oxygen (21%) and pure nitrogen(79%) are used. It is very easy to ignite the flame when the room air is used. Please explain the phenomenon.)

答：配的纯气体混合物中不含氢类物质，而室内空气为湿空气，就有可能大大加强CO初OH氧化的反应。

5 （15分）在火箭的燃烧室中含有H原子和OH游基, 其浓度均为$4\times10^{-6}mol/cm^3$,温度为3000K，总的气体浓度为$4\times10^{-4}mol/cm^3$。若这些气体是在1000K的温度下从燃烧室排出，此时H_2O基本上不离解, 密度是燃烧室内气体密度的1/40，若排气以3048m/s的速度流动,试计算在下游中多远的地方能测出H和OH已有99%再化合。假设除

H+OH+M→H_2O+M外(k=$10^{16}cm^6/mol^2s$,且它与温度无关), 不考虑其他的再化合反应。(There are H radicals and OH radicals in the combustor of rocket. The concentrations of H and OH are same as $4\times10^{-6}mol/cm^3$. The temperature of the combustor is 3000K. The total gas concentration is $4\times10^{-4}mol/cm^3$. If the gas exits the combustor at the temperature of 1000K the formed H_2O does not dissociate and the density of the exit gas is about 1/40 of the density of the gas in combustor and the gas velocity is 3048m/s. Please determine the distance of the gas flow from the exit to the point of the downstream where H and OH recombined to 99%. Assume that only the recombine reaction takes place: H+OH+M→H_2O+M and k=$10^{16}cm^6/mol^2s$ and k is independent of temperature.)

答：只考虑一个反应，则反应速度为

$$\frac{\mathrm{d}[\mathrm{H}]}{\mathrm{d}t}=-k[\mathrm{M}][\mathrm{OH}],$$

将[M]看作一个常数，即为$4\times10^{-4}mol/cm^3/40=10^{-5}$。同时[OH]=[H]。

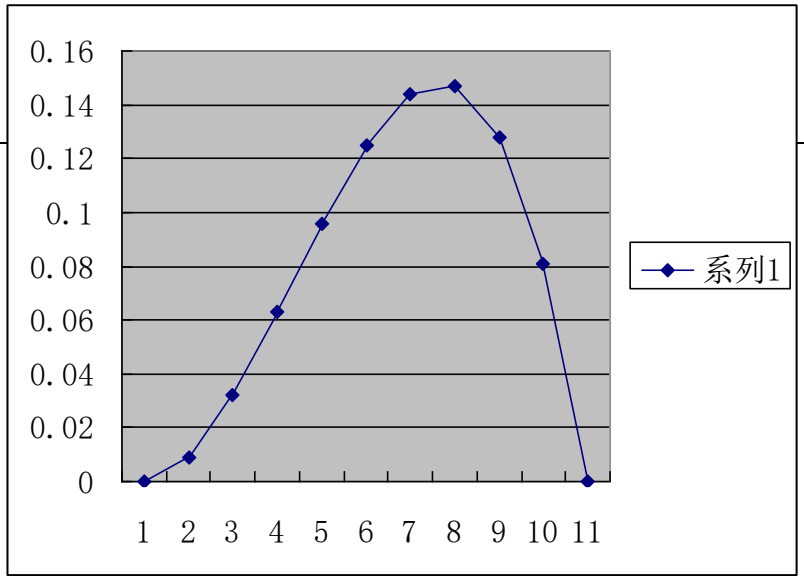
上式为$\frac{\mathrm{d}[\mathrm{H}]}{\mathrm{d}t}=-k[\mathrm{M}][\mathrm{H}]^2$，即$\frac{1}{[\mathrm{H}]_t}=\mathrm{k[M]t}+\frac{1}{[\mathrm{H}]_o}$，可计算出$t_{0.99}$=99/($4\times10^{-6}$/40*$10^{16}$*$10^{-5}$)

=0.0099秒，0.0099*3048，则距离为30.17米。

6 (10分)若三级反应αA+βB=γD的反应速度可表示为(A recombine reaction is third orderαA+βB=γD and its reaction rate can be expressed as) :

$$w=-\frac{\mathrm{d}C_A}{\mathrm{d}\tau}=kC_A^2C_B \qquad w=-\frac{\mathrm{d}[A]}{\mathrm{d}\tau}=k[A]^2[B]$$

且反应物A和B中均无惰性气体，试说明反应物A的相对浓度A1对反应速度*w*的影响情况，定性地画出变化曲线(若反应过程中温度与压力均保持不变),并求出反应速度达到最大值时相对浓度A1的值 。(There are no inert gases in the mixture. Please show the effect of relative concentration A1of reactant A on the reaction rate *w*. Draw the qualitative curve（Assume the temperature and pressure are the same in the reaction process. Determine the maximum value of A1.）



答：无其他气体，假设化学当量配比，A1=α/(α+β), B1=1-A1，代入上式可以有代表性，同时w可以对A1求导可以并令其为零可以获得合理的最大值。在图中标出就可以。合理的结果是A1=2/3时在到最大值。

7 (15分)一个直径25 mm的试管中装有水，在1 atm的压力下，水蒸发进入干空气中。已知气－水界面距试管口 $L=15$ cm，且界面上水蒸气的质量分数为0.0235，水蒸气相对空气的二元扩散率为 $2.6\cdot10^{-5}$ m^2/s。(There is water in a test tube with diameter of 25mm at the pressure of 1atm. The water evaporated into the dry air. The air-water surface is 15 cm lower from the tube tip. The mass fraction of water vapor is 0.0235 at the interface surface and the binary diffusivity of water vapor in air is $2.6\cdot10^{-5}$ m^2/s.

7.1 求水的质量蒸发速率。(Determine the mass evaporation rate)

7.2 求 $x=L/2$ 处水蒸气的质量分数。(determine the mass fraction of the vapor at $x=L/2$)

7.3 求在 $x=L/2$ 处因整体流动产生的水蒸汽质量通量的份额和因扩散产生的水蒸汽质量通量的份额。(Determine the parts of the water vapor mass flux for the bulk flow and for diffusion respectively)

答：这是标准的stefan 流问题。可以假设为常温，则可以直接用书上的结论。

7.1先求 $\dot{m}''_A=\frac{\rho D_{AB}}{L}\ln[\frac{1-Y_{A,\infty}}{1-Y_{A,i}}]$，L=15cm, $Y_{A,\infty}$=0，$Y_{A,i}$=0.0235,ρ=1.2,则计算得水的蒸发速率为

$$\dot{m}''_{H_2O}=\frac{1.2\left(2.6\cdot10^{-5}\right)}{0.15}\ln\left[\frac{1-0}{1-0.0235}\right]$$
$$=4.94635\cdot10^{-6}\ \text{kg/s-m}^2$$

$$\dot{m}_{H_2O}=\dot{m}''_{H_2O}\frac{\pi D^2}{4}=4.94635\cdot10^{-6}\frac{\pi\left(0.025\right)^2}{4}=\boxed{2.42803\cdot10^{-9}\ \text{kg/s}}$$

7.2:用式3.39有 $Y_A(x)=1-(1-Y_{A,i})\exp[\frac{\dot{m}''_A x}{\rho D_{AB}}]$

$$=1-(1-0.0235)\exp[\frac{4.94635*10^{-9}*0.075}{1.2*2.6*10^{-5}}]$$

=0.01182(还有一个简化的方法，假设线性分布，则为0.0235/2=0.01175)

7.3: 宏观流引起：$Y_A * m_A''$=0.01182*4.94635#10^{-6}=5.84651*10^{-8}kg/s-m^2

扩散流为$m_A'' - Y_A * m_A''$=4.94635*10^{-6}-5.84651*10^{-8}=4.88788*10^{-6} kg/s-m^2

注：宏观流只占1.182%,扩散占98.818%

8 (15分)燃料电池是一种无火焰的燃烧，其理论效率可表示为$\eta_{\text{th}} = \Delta G_T^0 / \Delta H_{298}^0$，$\Delta G_T^0$是工作温度下燃料电池反应的标准生成吉布斯函数的变化，ΔH_{298}^0则是标准参考状态下燃料电池反应标准生成焓的变化。如果采用H_2为燃料，则燃料电池的反应为，(Fuel cell is a kind of non-flame of combustion. The theoretic efficiency is $\eta_{\text{th}} = \Delta G_T^0 / \Delta H_{298}^0$. ΔG_T^0 is the standard Gibbs function change of the fuel cell reaction in the operation temperature. ΔH_{298}^0 is the standard enthalpy of formation change at the standard reference state. The fuel is Hydrogen and the fuel cell reaction is:)

$$H_2\ (g) + 1/2O_2\ (g) \rightarrow H_2O\ (l)$$

l, g分别表示液态和气态，则求解(Where,l,g are liquid and gas respectively)

A. 对于工作温度是298K的离子交换型燃料电池，试求η_{th}(Determine the η_{th} when the temperature is 298K. It is the PEMFC)

B. 对于工作温度是1000K的固体氧化物型燃料电池，试求η_{th}(Determine the η_{th} when the temperature is 1000K. It is the SOFC)

答：

这个题目看上去容易，做起来不易：

$$\Delta H_{298} = -241845 - 44010 = -285855$$

A. $\Delta G_{298,l} = -228608$(相变过程为等温、等压过程，所以G不变)

$$\eta_{th} = \Delta G_{298,l} / \Delta H_{298} = \frac{-228608}{-285855} = 81.889\%$$

B. $\Delta G_{1000,g} = -192652$

$$\eta_{th} = \Delta G_{1000,g} / \Delta H_{298} = \frac{-192652}{-285855} = 67.395\%$$